



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ3553 ARTIFICIAL INTELLIGENCE

SECTION 02

Assignment 3

LECTURER: DR. ROLIANA BINTI IBRAHIM

GROUP

Student Name	Matric No.
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220
GOE JIE YING	A23CS0224
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187
TEH RU QIAN	A23CS0191

1.0 Type of Agent

Our proposed application, **Rydit: Learn Mandarin** is a model-based reflex agent, where the model tracks its previous states. The agent tracks the accuracy of the past attempts to decide if a new character shall be introduced and the frequency of the character's appearance. The current percepts will be used to determine the accuracy. The current percepts along with the previous states, allow the agent to make more informed decisions about its actions.

2.0 PEAS Model

The table below shows the formulated solution by using the PEAS model.

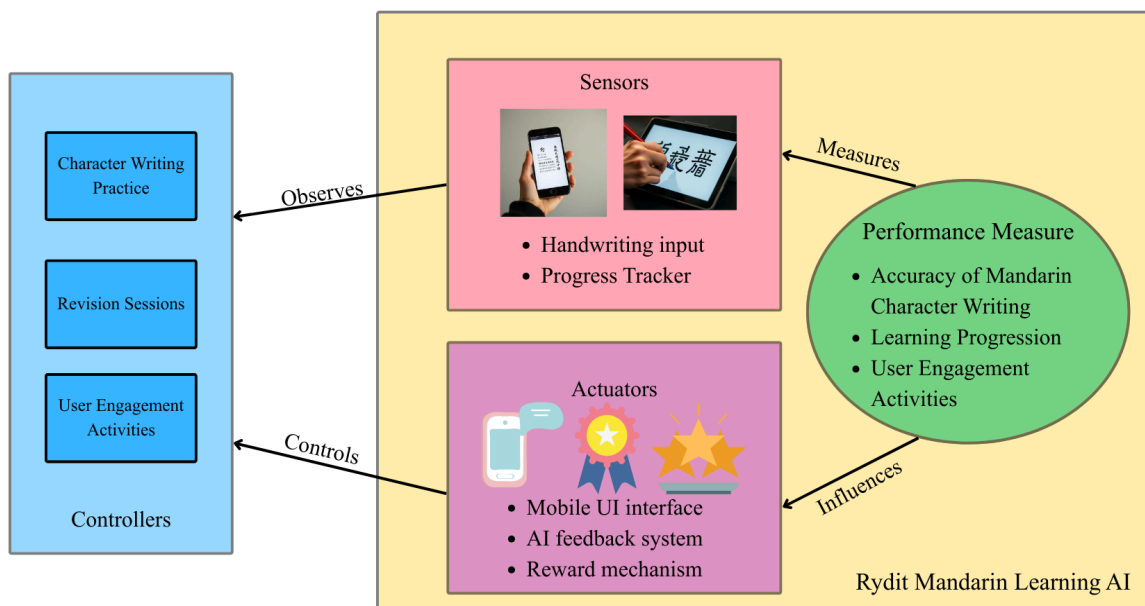
P: Performance Measure	Accuracy of Mandarin character writing, and ● Improvement in user handwriting accuracy (stroke order, direction, and similarity) ● Reduction in user error rate over time ● Successful mastery of characters ($\geq 90\%$ accuracy in last 5 attempts) ● Effective retention of characters through spaced repetition Learning progression ● Timely and appropriate introduction of new characters User engagement ● Consistent user engagement and learning streaks ● Positive user feedback and motivation levels (reward points, badges, progress)
E: Environment	Mobile application platform ● Mobile learning environment ● Touchscreen device (tablet or smartphone) ● Gamified elements such as rewards, points, and achievements Learners ● Mandarin learners with varying skill levels Learning sessions ● Learning sessions consisting of: ○ Character revision ○ New word introduction ○ Handwriting practice Mandarin learning context ● Dynamic learning context influenced by: ○ User accuracy ○ Error rate ○ Revision history ○ Learning frequency
A: Actuators/ Effectors	Mobile application interface ● Displaying characters for revision or learning ● Introducing new Mandarin characters ● Moving the user to the next learning question AI feedback system

	• Providing real-time handwriting feedback • Highlighting stroke order and direction corrections • Adjusting revision frequency using spaced repetition • Delivering personalised AI feedback (hints, corrections, encouragement) Reward mechanism • Awarding reward points, badges, and achievements • Triggering motivational messages and progress notifications
S: Sensor	Handwriting input data • User handwriting input (stroke order, direction, shape) User performance data • User answer correctness • Accuracy scores from recent attempts • Error rate per character • Revision interval data Learning progress indicators • Learning progress and mastery status • Frequency and consistency of user practice • User interaction patterns within the app

2.0 Relations Between Each Property

This diagram shows how the Rydit Mandarin learning app works:

- **Controllers** decide what activity to give next, such as writing practice or revision
- **Sensors** to observe users write and track their learning progress
- **Actuators** respond by giving feedback or rewards
- **Performance measures** to evaluate accuracy, progression and engagement.



3.0 Explanation In Proof Of Concept (POC)

3.1 Performance Measure

Our system is able to help users learn and memorize Mandarin characters through **smart rotation of questions** based on users' exercise patterns. When users are learning to write a character, the system can accurately identify errors when **evaluating user handwriting** based on stroke order, stroke direction and character similarity. The system is able to identify users' weak points and help them overcome even the hardest characters by providing **personalized AI feedback**. The system made learning mandarin painless by gradually increasing the vocabulary, through the **introduction of new words** while ensuring practices on every previously learnt word once in a while. The system can retain users' attention, maintain user engagement and interest in learning Mandarin by giving **rewards** upon goal completion.

3.2 Environment

The operating environment of the system consists of users ranging from 3 to 80 years old that interact with the system using mobile devices. The system operates with a mobile learning environment that uses touchscreen-based handwriting input for learning and practicing Mandarin characters. The environment provides structured learning methods including handwriting practice, character revision and the introduction of new vocabulary. The environment is dynamic and evolves based on user interaction patterns, learning frequency and historical performance data. The system integrates with game-based learning elements, such as reward mechanisms and achievement indicators, which guide the user's learning context and encourages interaction.

3.3 Actuators

The system operates through a mobile application. The mobile interface displays the characters for revision and new characters for learning. The mobile interface will guide the user through each task. Additionally, the AI feedback system provides real-time feedback on handwriting accuracy, including stroke order and direction corrections. After each learning cycle, the system delivers motivational messages and advances the user to the next character. Along with a reward mechanism that awards points, badges, and achievements when positive learning behaviour is detected.

3.4 Sensors

The primary sensor captures the user's handwriting input, including stroke order, stroke direction, and stroke shape. Besides, the system keeps track of the user performance data, including its accuracy and revision intervals for each character. In addition, the system tracks users' learning progress, including the mastery status and practice consistency.

4.0 Behaviours of Agent to Achieve the Goals

In our proposed **Rydit: Learn Mandarin**, the AI agent behaves as a model-based reflex agent. It continuously observes user actions, updates its internal model and takes appropriate action to assist the users in learning Mandarin and achieve their learning goals. Several agent behaviour are implemented in our solution:

1. Performance Monitoring Behaviour

The agent perceives user handwriting input through sensors, including the handwriting's stroke order, shape and direction. It also records user's recent attempts to extract their error rates and accuracy scores. This is to understand the user's learning progress and maintain an updated internal state in order to provide the most appropriate learning material for the user.

2. Error Detection and Feedback Behaviour

When incorrect handwriting is detected, the agent provides real-time feedback by highlighting the mistakes and offering corrective guidance for the user. The personalized feedback can help the users to know their errors and improve writing accuracy for next attempts.

3. Adaptive Content Selection Behaviour

By referencing the stored learning history, the agent decides to introduce new Mandarin characters when mastery is detected and repeat the characters if the error rate is high. The agent controls and selects the learning content that most suits the users to ensure a balanced learning pace and improve users' language skill effectively.

4. Motivation and Engagement Behaviour

To encourage continuous learning and engagement, the agent triggers awards, badges and positive feedback if the users achieve certain milestones while learning. This includes maintaining a learning streak, showing progress, solving difficulties and reaching a high accuracy when learning.